Baboon Teeth Scott Scores

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Intro

Baboon teeth

Questions from Department Chair Krueger

Lastly, here are the questions: - Do wear scores increase with age? (Yes, duh, but we want to test it since we modified the method) - Do wear scores differ by sex? If so, how? (Is this by a specific quadrant? Is this by a side of tooth like buccal side or lingual side?) - Are there any patterns for rate of wear? That is, is there a standard wear through time (by month or year), or are some baboons wearing their teeth down faster that what we might expect?

Exploratory Data Analysis

```
Warning: NAs introduced by coercion
```

Summary statistics

```
library(tidyverse)
scott <- scott %>% mutate(diff = age_death - age_cast)

#by sex
scott %>% group_by(sex) %>%
  summarize(n = n(),
            mean_age_cast = mean(age_cast),
            mean_age_death = mean(age_death),
            mean_diff = mean(diff)) %>% as.data.frame()
```

```
     sex   n mean_age_cast mean_age_death mean_diff
1 female 150      233.5333       262.9600  29.42667
2   male  51      196.0980       224.3725  28.27451
```

```
#overall
scott  %>%
  summarize(n = n(),
            mean_age_cast = mean(age_cast),
            mean_age_death = mean(age_death),
            mean_diff = mean(diff))
```

```
    n mean_age_cast mean_age_death mean_diff
1 201      224.0348       253.1692  29.13433
```

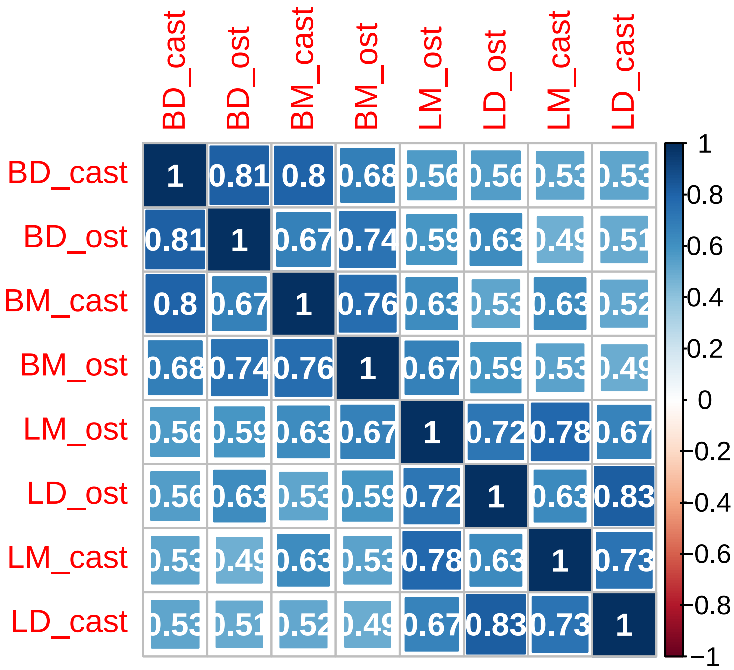
Variable summaries

```
summary(scott)
```

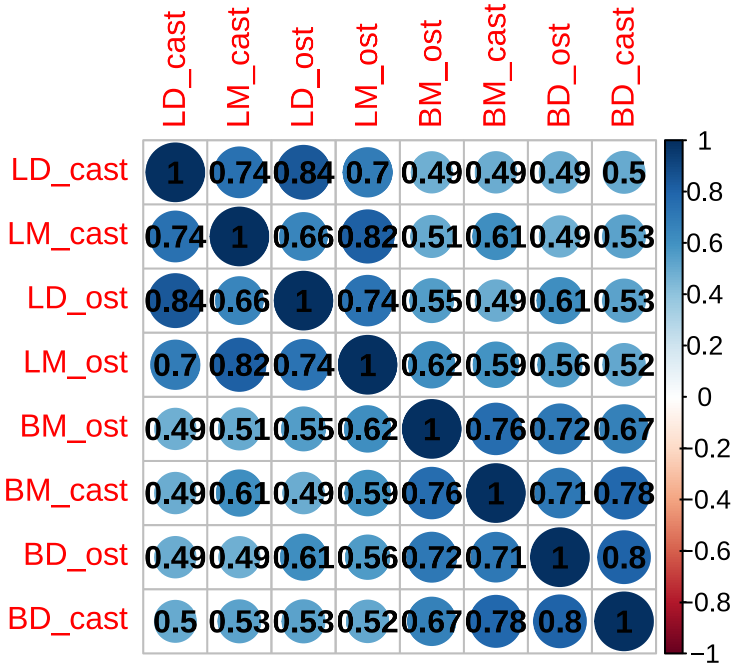
```
       id             sex               age_cast     age_death    
 Min.   :  3.0   Length:201         Min.   : 85   Min.   :102.0  
 1st Qu.:171.0   Class :character   1st Qu.:185   1st Qu.:215.0  
 Median :264.0   Mode  :character   Median :232   Median :260.0  
 Mean   :285.6                      Mean   :224   Mean   :253.2  
 3rd Qu.:420.0                      3rd Qu.:256   3rd Qu.:289.0  
 Max.   :621.0                      Max.   :401   Max.   :404.0  
    sfbrid           leftright            BM_cast         BD_cast     
 Length:201         Length:201         Min.   :2.000   Min.   :1.000  
 Class :character   Class :character   1st Qu.:5.000   1st Qu.:5.000  
 Mode  :character   Mode  :character   Median :6.000   Median :5.000  
                                       Mean   :5.851   Mean   :5.308  
                                       3rd Qu.:6.000   3rd Qu.:6.000  
                                       Max.   :9.000   Max.   :9.000  
    LM_cast        LD_cast          BM_ost          BD_ost         LM_ost     
 Min.   :1.00   Min.   :1.000   Min.   :4.000   Min.   :1.00   Min.   :2.000  
 1st Qu.:2.00   1st Qu.:2.000   1st Qu.:6.000   1st Qu.:5.00   1st Qu.:4.000  
 Median :4.00   Median :4.000   Median :6.000   Median :6.00   Median :5.000  
 Mean   :4.09   Mean   :3.622   Mean   :6.502   Mean   :6.06   Mean   :5.348  
 3rd Qu.:5.00   3rd Qu.:5.000   3rd Qu.:7.000   3rd Qu.:7.00   3rd Qu.:6.000  
 Max.   :9.00   Max.   :9.000   Max.   :9.000   Max.   :9.00   Max.   :9.000  
     LD_ost           diff      
```

```
 Min.   :1.000   Min.   : 1.00
 1st Qu.:3.000   1st Qu.:15.00
 Median :5.000   Median :29.00
 Mean   :4.687   Mean   :29.13
 3rd Qu.:6.000   3rd Qu.:43.00
 Max.   :9.000   Max.   :59.00
```

Correlations



Spearman correlations



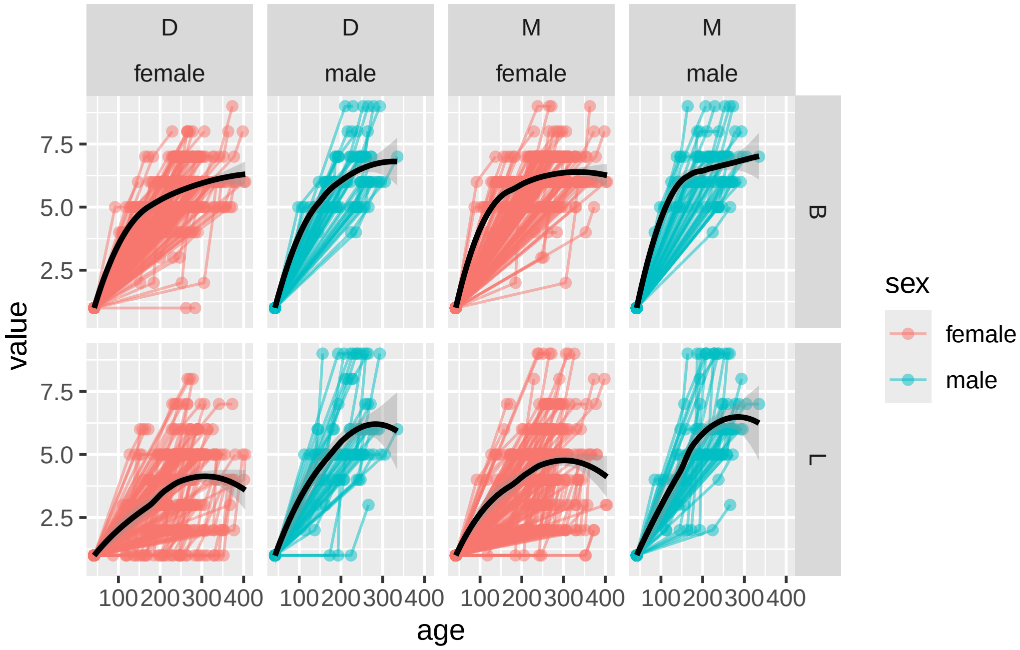
P-values for the Spearman correlation test

	BM_cast	BD_cast	LM_cast	LD_cast	BM_ost
BM_cast	NA	1.513686e-42	2.872028e-22	1.529116e-13	7.478033e-40
BD_cast	NA	NA	5.747780e-16	2.357984e-14	6.081874e-28
LM_cast	NA	NA	NA	9.839248e-37	1.634475e-14
LD_cast	NA	NA	NA	NA	2.202239e-13
BM_ost	NA	NA	NA	NA	NA
BD_ost	NA	NA	NA	NA	NA
LM_ost	NA	NA	NA	NA	NA
LD_ost	NA	NA	NA	NA	NA

	BD_ost	LM_ost	LD_ost
BM_cast	3.032539e-32	1.418489e-20	1.157733e-13
BD_cast	6.837721e-47	3.679490e-15	4.534340e-16
LM_cast	2.431100e-13	1.678756e-49	2.500371e-26
LD_cast	1.504922e-13	6.651144e-31	5.686140e-55
BM_ost	2.609605e-33	1.056400e-22	1.370442e-17
BD_ost	NA	2.924310e-18	5.951075e-22
LM_ost	NA	NA	1.477813e-35
LD_ost	NA	NA	NA

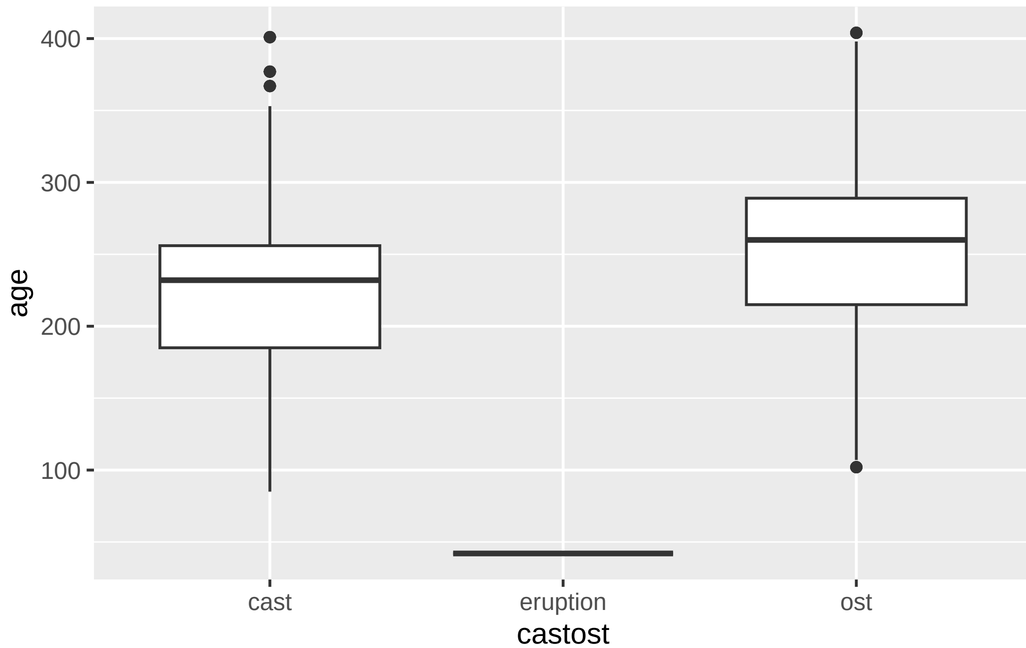
The Bonferroni corrected cut off here is 0.0017857. All of these p-values are well below this cut-off.

Data Viz: Age vs wear by sex



Age distributions by measurement time.

```
ggplot(aes(x = castost, y = age), data = scott_long %>% filter(age > 1)) + geom_boxplot() +
```



Do wear scores increase with age? (Yes, duh, but we want to test it since we modified the method)

The tests below test the following hypothesis: $H_0 : \mu_{cast} = \mu_{ost}$ vs $H_0 : \mu_{cast} < \mu_{ost}$ for each location of the tooth individually. The test used here is a two sample dependent t-test. In all cases, the null hypothesis is rejected with a p-value less than 2.2×10^{-16}. There is statistical evidence that the mean at ost is larger than the mean at cast.

```
t.test(scott$BM_cast, scott$BM_ost, paired = TRUE, alternative = "less")
```

```

    Paired t-test

data:  scott$BM_cast and scott$BM_ost
t = -13.803, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
       -Inf -0.5737156
sample estimates:
mean difference
     -0.6517413
```

```r
t.test(scott$BD_cast, scott$BD_ost, paired = TRUE, alternative = "less")
```

```
    Paired t-test

data:  scott$BD_cast and scott$BD_ost
t = -15.408, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
       -Inf -0.6706737
sample estimates:
mean difference
     -0.7512438
```

```r
t.test(scott$LM_cast, scott$LM_ost, paired = TRUE, alternative = "less")
```

```
    Paired t-test

data:  scott$LM_cast and scott$LM_ost
t = -15.343, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf -1.12314
sample estimates:
mean difference
      -1.258706
```

```r
t.test(scott$LD_cast, scott$LD_ost, paired = TRUE, alternative = "less")
```

```
    Paired t-test

data:  scott$LD_cast and scott$LD_ost
t = -14.071, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
       -Inf -0.9396378
sample estimates:
mean difference
      -1.064677
```

No matter what correction you use here, these are going to be highly significant.

Modeling effect of age

Here are pearson correlation values for you. But the relationship clearly isn't linear so these probably aren't what you want. I've also computed Spearman correlation for you, which tests for any kind of monotonic relationship. This is much closer to what you are looking for than Pearson's coefficient.

Pearson correlation tests

I would argue there are 8 tests here in this family so the Bonferroni corrected cut off would be 0.00625.

```
#Pearson correlation
cor.test(scott$BM_cast,scott$age_cast)
```

```

    Pearson's product-moment correlation

data:  scott$BM_cast and scott$age_cast
t = 3.1426, df = 199, p-value = 0.00193
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.08150242 0.34544369
sample estimates:
      cor
0.2174445
```

```
cor.test(scott$BD_cast,scott$age_cast)
```

```

    Pearson's product-moment correlation

data:  scott$BD_cast and scott$age_cast
t = 4.6134, df = 199, p-value = 7.086e-06
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1801893 0.4306997
sample estimates:
```

```
      cor
0.3108327
```

```r
cor.test(scott$LM_cast,scott$age_cast)
```

```

	Pearson's product-moment correlation

data:  scott$LM_cast and scott$age_cast
t = 1.5346, df = 199, p-value = 0.1265
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.03070694  0.24290650
sample estimates:
      cor
0.1081474
```

```r
cor.test(scott$LD_cast,scott$age_cast)
```

```

	Pearson's product-moment correlation

data:  scott$LD_cast and scott$age_cast
t = 1.258, df = 199, p-value = 0.2098
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.05018341  0.22446354
sample estimates:
       cor
0.08882823
```

```r
cor.test(scott$BM_ost,scott$age_death)
```

```

	Pearson's product-moment correlation

data:  scott$BM_ost and scott$age_death
t = 2.2123, df = 199, p-value = 0.02808
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

```
 0.01690182 0.28717130
sample estimates:
      cor
0.1549341
```

```r
cor.test(scott$BD_ost,scott$age_death)
```

```

    Pearson's product-moment correlation

data:  scott$BD_ost and scott$age_death
t = 4.0452, df = 199, p-value = 7.477e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1426931 0.3988249
sample estimates:
      cor
0.2756444
```

```r
cor.test(scott$LM_ost,scott$age_death)
```

```

    Pearson's product-moment correlation

data:  scott$LM_ost and scott$age_death
t = 0.99168, df = 199, p-value = 0.3226
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.06893834  0.20651590
sample estimates:
       cor
0.07012542
```

```r
cor.test(scott$LD_ost,scott$age_death)
```

```

    Pearson's product-moment correlation

data:  scott$LD_ost and scott$age_death
t = 1.1993, df = 199, p-value = 0.2318
```

```
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.0543204  0.2205206
sample estimates:
       cor
0.08471114
```

Spearman correlation tests

```r
#Spearman correlation
cor.test(scott$BM_cast,scott$age_cast, method = "spearman")
```

```
    Spearman's rank correlation rho

data:  scott$BM_cast and scott$age_cast
S = 970966, p-value = 4.817e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
     rho
0.282573
```

```r
cor.test(scott$BD_cast,scott$age_cast, method = "spearman")
```

```
    Spearman's rank correlation rho

data:  scott$BD_cast and scott$age_cast
S = 875455, p-value = 2.717e-07
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.3531437
```

```r
cor.test(scott$LM_cast,scott$age_cast, method = "spearman")
```

```
    Spearman's rank correlation rho
```

```
data:  scott$LM_cast and scott$age_cast
S = 1164898, p-value = 0.04862
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1392802
```

```r
cor.test(scott$LD_cast,scott$age_cast, method = "spearman")
```

```
    Spearman's rank correlation rho

data:  scott$LD_cast and scott$age_cast
S = 1211368, p-value = 0.1382
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1049448
```

```r
#Corelation
cor.test(scott$BM_ost,scott$age_death, method = "spearman")
```

```
    Spearman's rank correlation rho

data:  scott$BM_ost and scott$age_death
S = 1146997, p-value = 0.03067
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1525068
```

```r
cor.test(scott$BD_ost,scott$age_death, method = "spearman")
```

```
    Spearman's rank correlation rho

data:  scott$BD_ost and scott$age_death
S = 946152, p-value = 1.421e-05
alternative hypothesis: true rho is not equal to 0
```

```
sample estimates:
      rho
0.3009077
```

```r
cor.test(scott$LM_ost,scott$age_death, method = "spearman")
```

```

    Spearman's rank correlation rho

data:  scott$LM_ost and scott$age_death
S = 1245286, p-value = 0.2596
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.07988334
```

```r
cor.test(scott$LD_ost,scott$age_death, method = "spearman")
```

```

    Spearman's rank correlation rho

data:  scott$LD_ost and scott$age_death
S = 1213542, p-value = 0.1443
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1033379
```

P-values for relationship between wear and age.

I would argue there are 8 tests here in this family so the Bonferroni corrected cut off would be 0.00625. Using the cut-off, the tests that are significant are for BM cast and age, BD cast and age, and BD ost and age.

```r
out %>% pivot_wider(names_from = "BDLM", values_from = "pvalue")
```

```
# A tibble: 2 x 5
  castost         BM          BD     LM    LD
  <chr>        <dbl>       <dbl>  <dbl> <dbl>
1 cast     0.0000482 0.000000272 0.0486 0.138
2 ost      0.0307    0.0000142   0.260  0.144
```

Do wear scores differ by sex? If so, how? (Is this by a specific quadrant? Is this by a side of tooth like buccal side or lingual side?)

If we ignore age entirely, there are significant differences in the mean vector of wear for males vs females at bost cast and ost. If we then do individual tests for wear differences by region at cast and ost separately, there are signficant differences in LM and LD at cast and LM at ost. (I used a 0.05 level of significant and a Bonferroni correction with 8 tests to gets a cut off for significance of 0.00625. This is, as always, conservative.)

```
#Look for differences in male vs female with a hotelling t test
library(Hotelling)
```

```
Loading required package: corpcor
```

```
Attaching package: 'Hotelling'
```

```
The following object is masked from 'package:dplyr':

    summarise
```

```
results_cast <- hotelling.test(.~sex, data = scott %>% select(sex,BM_cast, BD_cast, LM_cast,
results_cast
```

```
Test stat:  42.938
Numerator df:  4
Denominator df:  86.9802290794176
P-value:  4.999e-07
```

```
results_ost <- hotelling.test(.~sex, data = scott %>% select(sex,BM_ost, BD_ost, LM_ost, LD_
results_ost
```

```
Test stat:  46.791
Numerator df:  4
Denominator df:  77.4160522734112
P-value:  2.173e-07
```

```r
#Now do individual t-tests on each component
t.test(BM_cast~sex,data = scott)
```

```

	Welch Two Sample t-test

data:  BM_cast by sex
t = -1.118, df = 89.591, p-value = 0.2665
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -0.4824452  0.1349942
sample estimates:
mean in group female   mean in group male 
            5.806667             5.980392 
```

```r
t.test(BD_cast~sex,data = scott)
```

```

	Welch Two Sample t-test

data:  BD_cast by sex
t = -2.2027, df = 88.72, p-value = 0.03021
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -0.76308075 -0.03927219
sample estimates:
mean in group female   mean in group male 
            5.206667             5.607843 
```

```r
t.test(LD_cast~sex,data = scott)
```

```

	Welch Two Sample t-test

data:  LD_cast by sex
t = -5.8849, df = 82.279, p-value = 8.303e-08
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -2.259947 -1.118092
sample estimates:
```

```
mean in group female   mean in group male
            3.193333             4.882353
```

```r
t.test(LM_cast~sex,data = scott)
```

```
    Welch Two Sample t-test

data:  LM_cast by sex
t = -4.4302, df = 92.159, p-value = 2.589e-05
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -1.728868 -0.658583
sample estimates:
mean in group female   mean in group male
            3.786667             4.980392
```

```r
#Now do individual t-tests on each component for ost
t.test(BM_ost~sex,data = scott)
```

```
    Welch Two Sample t-test

data:  BM_ost by sex
t = -2.2186, df = 73.084, p-value = 0.02962
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -0.71688510 -0.03840902
sample estimates:
mean in group female   mean in group male
            6.406667             6.784314
```

```r
t.test(BD_ost~sex,data = scott)
```

```
    Welch Two Sample t-test

data:  BD_ost by sex
t = -2.7012, df = 78.207, p-value = 0.008468
alternative hypothesis: true difference in means between group female and group male is not e
```

```
95 percent confidence interval:
 -0.8650942 -0.1309843
sample estimates:
mean in group female   mean in group male
            5.933333             6.431373
```

```r
t.test(LD_ost~sex,data = scott)
```

```
	Welch Two Sample t-test

data:  LD_ost by sex
t = -6.2787, df = 78.517, p-value = 1.755e-08
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -2.318006 -1.201994
sample estimates:
mean in group female   mean in group male
                4.24                 6.00
```

```r
t.test(LM_ost~sex,data = scott)
```

```
	Welch Two Sample t-test

data:  LM_ost by sex
t = -4.6294, df = 76.952, p-value = 1.462e-05
alternative hypothesis: true difference in means between group female and group male is not e
95 percent confidence interval:
 -1.8877776 -0.7522224
sample estimates:
mean in group female   mean in group male
            5.013333             6.333333
```

Modeling

Scott Scores

But we should really be controlling for sex and age and id all at once and I think we should be looking at age at eruption with wear of 0.

Need to uses REML = FALSE to compare model. Reason why: -https://stats.stackexchange.com/questions/2726 to-decide-whether-to-set-reml-to-true-or-false/272654#272654 - https://stackoverflow.com/questions/54980399/ should-i-specify-reml-false-in-lmer

Scott Score overall

```
library(lme4)
```

```
Loading required package: Matrix


Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(va
```

```
`summarise()` has grouped output by 'id', 'castost', 'age'. You can override
using the `.groups` argument.
```

```
mod_all_full <- lmer(value ~ (age+I(age^2) + I(age^3))*sex + (1|id), data = scott_long_sum, R
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
mod_all_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long_sum, REML =
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
#test stat
-2*(logLik(mod_all_red) - logLik(mod_all_full))
```

```
'log Lik.' 103.8104 (df=6)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_all_red) - logLik(mod_all_full)),4)
```

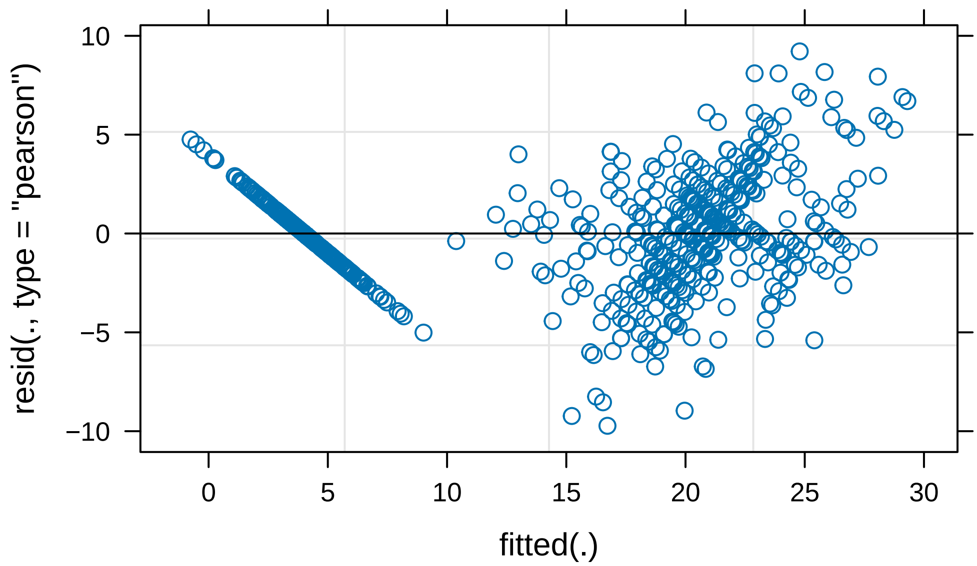
```
'log Lik.' 0 (df=6)
```

```r
summary(mod_all_full)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
   Data: scott_long_sum

     AIC      BIC   logLik deviance df.resid
  3250.6   3294.7  -1615.3   3230.6      593

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2130 -0.5006 -0.0202  0.5130  3.0428

Random effects:
 Groups   Name        Variance Std.Dev.
 id       (Intercept) 4.559    2.135
 Residual             9.164    3.027
Number of obs: 603, groups:  id, 201

Fixed effects:
                   Estimate Std. Error t value
(Intercept)      -3.513e+00  9.397e-01  -3.738
age               2.020e-01  2.559e-02   7.894
I(age^2)         -5.553e-04  1.500e-04  -3.703
I(age^3)          5.223e-07  2.448e-07   2.134
sexmale          -3.662e+00  2.178e+00  -1.681
age:sexmale       1.043e-01  6.333e-02   1.648
I(age^2):sexmale -4.483e-04  4.261e-04  -1.052
I(age^3):sexmale  6.910e-07  8.195e-07   0.843

Correlation of Fixed Effects:
            (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age         -0.943
I(age^2)     0.912 -0.991
I(age^3)    -0.879  0.967 -0.992
sexmale     -0.431  0.407 -0.394  0.379
```

```
age:sexmale  0.381 -0.404  0.400 -0.391 -0.957
I(g^2):sxml -0.321  0.349 -0.352  0.349  0.925 -0.989
I(g^3):sxml  0.263 -0.289  0.296 -0.299 -0.886  0.961 -0.990
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

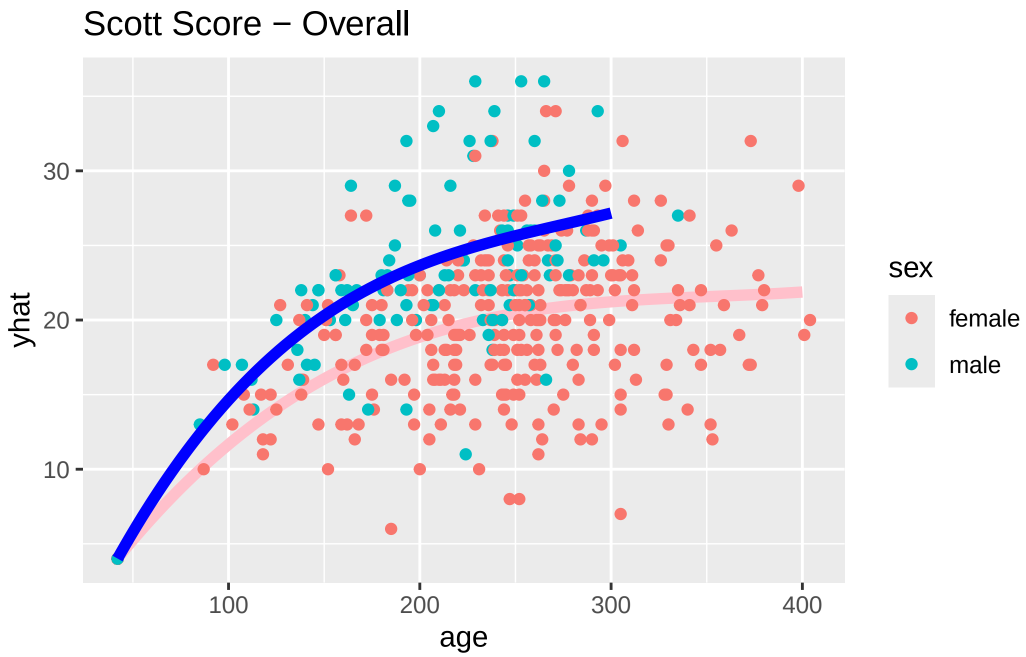
```
plot(mod_all_full)
```




```
confint(mod_all_full)
```

```
Computing profile confidence intervals ...

                         2.5 %        97.5 %
.sig01            1.774347e+00  2.516106e+00
.sigma            2.828582e+00  3.250379e+00
(Intercept)      -5.361139e+00 -1.664398e+00
age               1.516552e-01  2.523818e-01
I(age^2)         -8.501675e-04 -2.603172e-04
I(age^3)          4.124907e-08  1.003273e-06
sexmale          -7.938213e+00  6.156152e-01
age:sexmale      -2.001948e-02  2.286823e-01
I(age^2):sexmale -1.284882e-03  3.883343e-04
I(age^3):sexmale -9.184339e-07  2.300077e-06
```

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_all_full, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_all_full, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(a
```



BM model

```
library(lme4)
mod_bm <- lmer(value ~ (age+I(age^2) + I(age^3))*sex + (1|id), data = scott_long %>% filter(B
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
mod_bm_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BM
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
#test stat
-2*(logLik(mod_bm_red) - logLik(mod_bm))
```

```
'log Lik.' 27.50309 (df=6)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_bm_red) - logLik(mod_bm)),4)
```

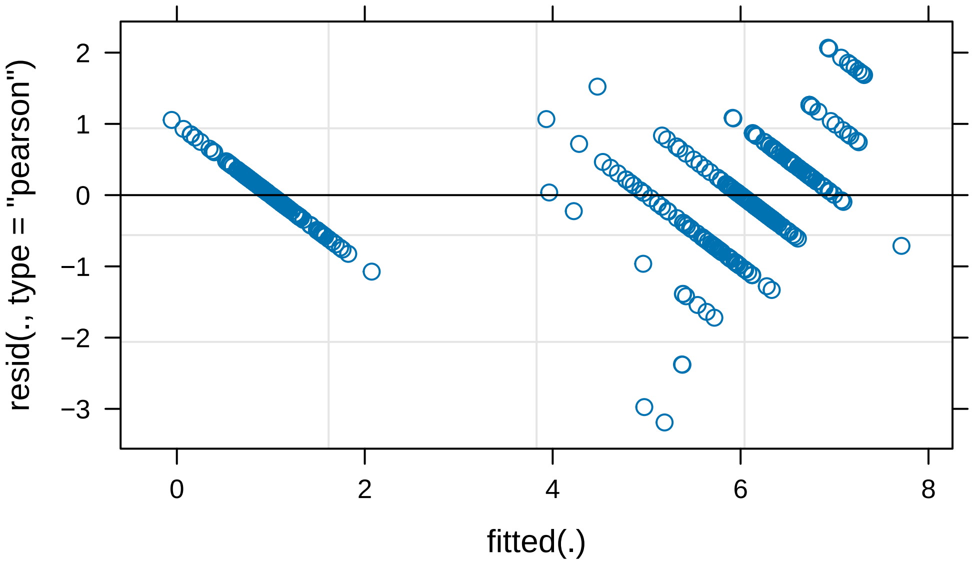
```
'log Lik.' 1.572599e-05 (df=6)
```

```r
summary(mod_bm)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
   Data: scott_long %>% filter(BLMD == "BM")

     AIC      BIC   logLik deviance df.resid
  1397.2   1441.2   -688.6   1377.2      593

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.8569 -0.4545 -0.0533  0.4862  3.1491

Random effects:
 Groups   Name        Variance Std.Dev.
 id       (Intercept) 0.1958   0.4424
 Residual             0.4316   0.6570
Number of obs: 603, groups:  id, 201

Fixed effects:
                   Estimate Std. Error t value
(Intercept)      -2.266e+00  2.021e-01 -11.209
age               9.115e-02  5.502e-03  16.567
I(age^2)         -3.248e-04  3.223e-05 -10.079
I(age^3)          3.857e-07  5.259e-08   7.334
sexmale          -1.232e+00  4.690e-01  -2.627
age:sexmale       3.834e-02  1.363e-02   2.813
I(age^2):sexmale -2.418e-04  9.168e-05  -2.638
I(age^3):sexmale  4.555e-07  1.764e-07   2.583
```

```
Correlation of Fixed Effects:
             (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age          -0.944
I(age^2)      0.913 -0.991
I(age^3)     -0.879  0.967 -0.992
sexmale      -0.431  0.407 -0.393  0.379
age:sexmale   0.381 -0.404  0.400 -0.390 -0.957
I(g^2):sxml  -0.321  0.348 -0.352  0.349  0.925 -0.989
I(g^3):sxml   0.262 -0.288  0.296 -0.298 -0.886  0.961 -0.990
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

```r
plot(mod_bm)
```

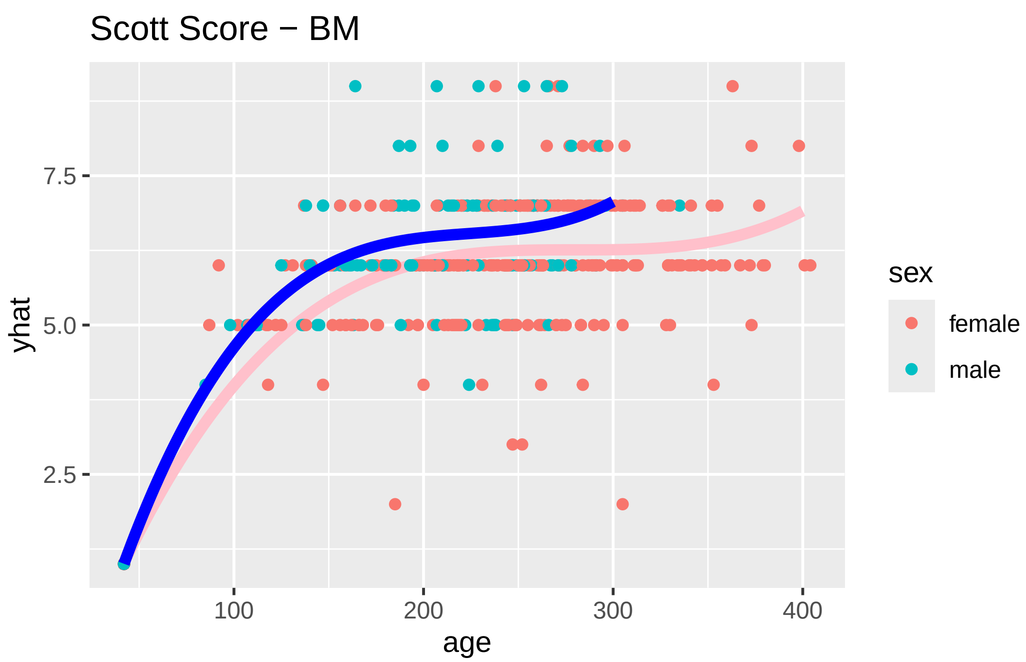



```r
confint(mod_bm)
```

```
Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation
```

```
                          2.5 %        97.5 %
.sig01             3.641796e-01  5.241933e-01
.sigma             6.138803e-01  7.053054e-01
(Intercept)       -2.663009e+00 -1.868469e+00
age                8.033269e-02  1.019677e-01
I(age^2)          -3.881395e-04 -2.614562e-04
I(age^3)           2.823800e-07  4.890441e-07
sexmale           -2.152828e+00 -3.115144e-01
age:sexmale        1.158228e-02  6.509252e-02
I(age^2):sexmale -4.217848e-04 -6.183189e-05
I(age^3):sexmale  1.092840e-07  8.017311e-07
```

```r
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_bm, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_bm, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(a
```



BD Model

```r
library(lme4)
mod_bd <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(Bl
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
mod_bd_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(Bl
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
#Test stat
-2*(logLik(mod_bd_red) - logLik(mod_bd))
```

```
'log Lik.' 50.48236 (df=6)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_bd_red) - logLik(mod_bd)),4)
```

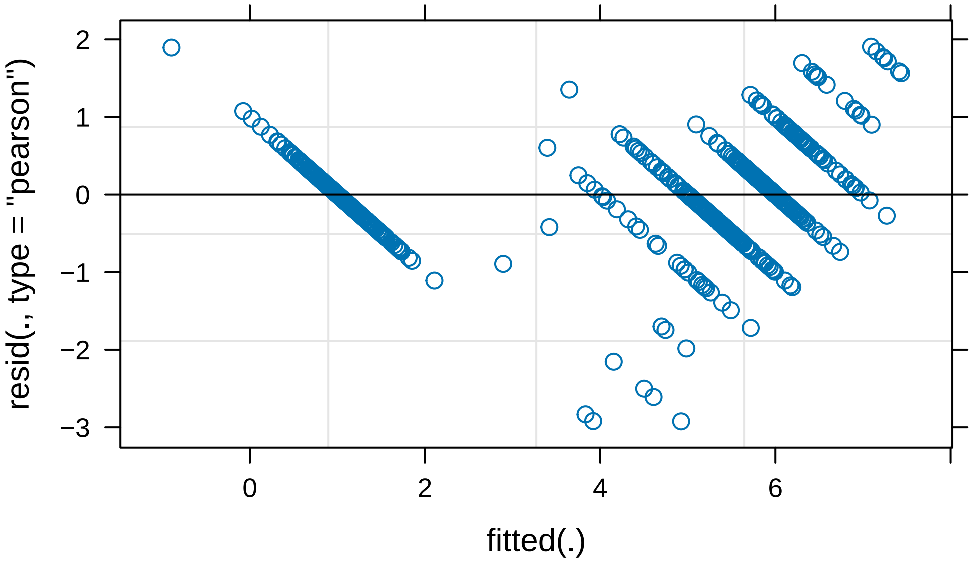
```
'log Lik.' 2.863381e-10 (df=6)
```

```r
summary(mod_bd)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
   Data: scott_long %>% filter(BLMD == "BD")

     AIC      BIC   logLik deviance df.resid
  1483.8   1527.8   -731.9   1463.8      593

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.1758 -0.4768  0.0364  0.4429  2.7227

Random effects:
 Groups   Name        Variance Std.Dev.
```

```
 id       (Intercept) 0.2417   0.4916
 Residual             0.4901   0.7001
Number of obs: 603, groups:  id, 201

Fixed effects:
                   Estimate Std. Error t value
(Intercept)      -1.591e+00  2.171e-01  -7.328
age               7.152e-02  5.913e-03  12.096
I(age^2)         -2.417e-04  3.465e-05  -6.975
I(age^3)          2.879e-07  5.656e-08   5.090
sexmale          -6.894e-01  5.034e-01  -1.370
age:sexmale       2.054e-02  1.463e-02   1.404
I(age^2):sexmale -1.078e-04  9.845e-05  -1.095
I(age^3):sexmale  1.982e-07  1.894e-07   1.046

Correlation of Fixed Effects:
            (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age         -0.944
I(age^2)     0.913 -0.991
I(age^3)    -0.879  0.967 -0.992
sexmale     -0.431  0.407 -0.394  0.379
age:sexmale  0.381 -0.404  0.400 -0.391 -0.957
I(g^2):sxml -0.321  0.349 -0.352  0.349  0.925 -0.989
I(g^3):sxml  0.263 -0.289  0.296 -0.299 -0.886  0.961 -0.990
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

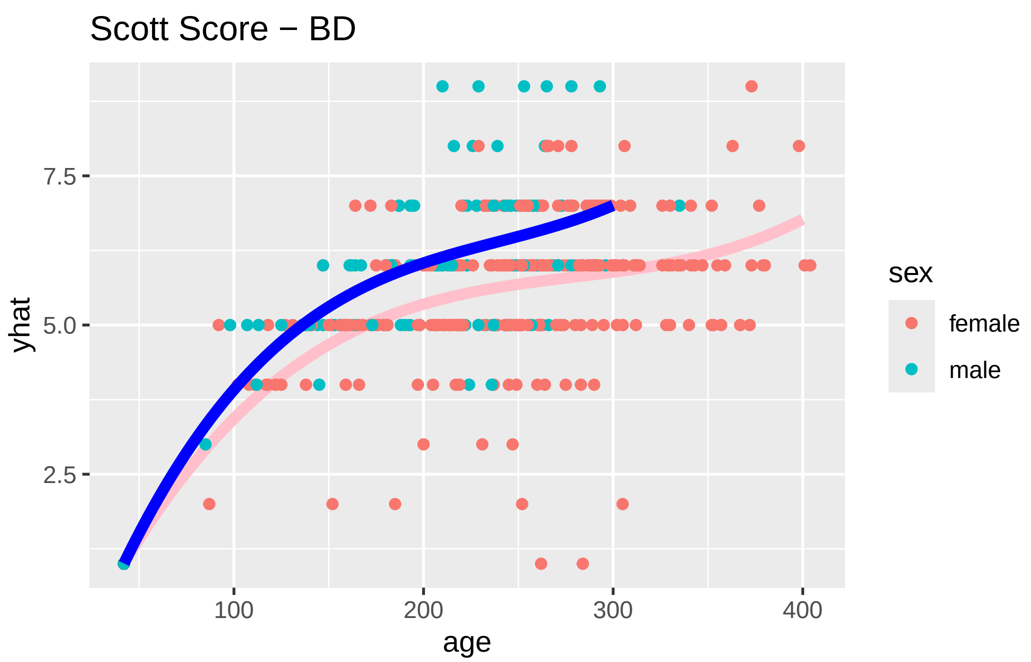
```r
plot(mod_bd)
```




```r
confint(mod_bm)
```

```
Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation

                         2.5 %        97.5 %
.sig01            3.641796e-01  5.241933e-01
.sigma            6.138803e-01  7.053054e-01
(Intercept)      -2.663009e+00 -1.868469e+00
age               8.033269e-02  1.019677e-01
I(age^2)         -3.881395e-04 -2.614562e-04
I(age^3)          2.823800e-07  4.890441e-07
sexmale          -2.152828e+00 -3.115144e-01
age:sexmale       1.158228e-02  6.509252e-02
I(age^2):sexmale -4.217848e-04 -6.183189e-05
I(age^3):sexmale  1.092840e-07  8.017311e-07
```

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_bd, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_bd, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(a
```



LM model

```
library(lme4)
mod_lm <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(Bl
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
mod_lm_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(Bl
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
#test stat
-2*(logLik(mod_lm_red) - logLik(mod_lm))
```

```
'log Lik.' 80.88223 (df=6)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_lm_red) - logLik(mod_lm)),4)
```

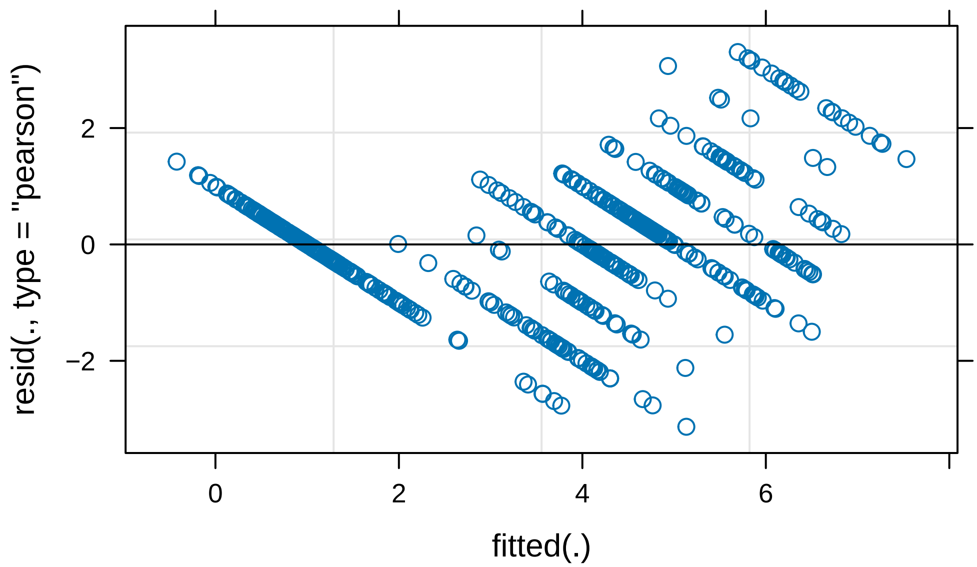
```
'log Lik.' 1.110223e-16 (df=6)
```

```r
summary(mod_lm)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
   Data: scott_long %>% filter(BLMD == "LM")

     AIC      BIC   logLik deviance df.resid
  2070.1   2114.1  -1025.0   2050.1      593

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.71290 -0.46049  0.02788  0.46279  2.86228

Random effects:
 Groups   Name        Variance Std.Dev.
 id       (Intercept) 0.566    0.7523
 Residual             1.334    1.1550
Number of obs: 603, groups:  id, 201

Fixed effects:
                   Estimate Std. Error t value
(Intercept)      -1.146e-01  3.532e-01  -0.324
age               2.787e-02  9.608e-03   2.901
I(age^2)         -2.609e-05  5.626e-05  -0.464
I(age^3)         -3.895e-08  9.180e-08  -0.424
sexmale          -6.130e-01  8.199e-01  -0.748
age:sexmale       1.524e-02  2.382e-02   0.640
I(age^2):sexmale -1.717e-05  1.602e-04  -0.107
I(age^3):sexmale -2.121e-08  3.082e-07  -0.069
```

```
Correlation of Fixed Effects:
             (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age          -0.945
I(age^2)      0.913 -0.991
I(age^3)     -0.879  0.967 -0.992
sexmale      -0.431  0.407 -0.393  0.379
age:sexmale   0.381 -0.403  0.400 -0.390 -0.958
I(g^2):sxml -0.321  0.348 -0.351  0.348  0.925 -0.989
I(g^3):sxml  0.262 -0.288  0.295 -0.298 -0.886  0.961 -0.990
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

```
plot(mod_lm)
```

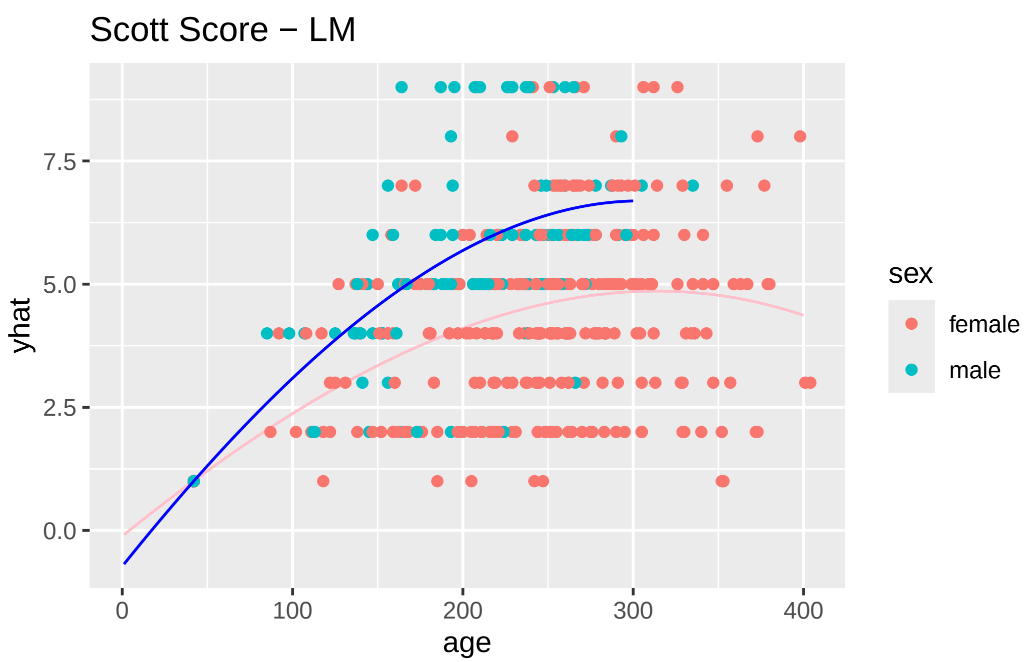



```
confint(mod_lm)
```

```
Computing profile confidence intervals ...

                       2.5 %       97.5 %
.sig01           6.138133e-01 8.956860e-01
.sigma           1.079227e+00 1.240107e+00
(Intercept)     -8.098491e-01 5.806864e-01
age              8.940271e-03 4.679066e-02
I(age^2)        -1.367742e-04 8.468194e-05
I(age^3)        -2.195181e-07 1.414884e-07
```

```
sexmale           -2.222479e+00 9.966918e-01
age:sexmale       -3.151813e-02 6.200393e-02
I(age^2):sexmale -3.316920e-04 2.973565e-04
I(age^3):sexmale -6.263407e-07 5.838899e-07
```

```
newx <- data.frame(age = 1:400, sex = "female")
newx$yhat <- (predict(mod_lm, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 1:300, sex = "male")
newxm$yhat <- (predict(mod_lm, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink") + geom_point(aes(x = ag
```



LD Model

```
library(lme4)
mod_ld <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(BI
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
mod_ld_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BL
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```r
#Test stat
-2*(logLik(mod_ld_red) - logLik(mod_ld))
```

```
'log Lik.' 136.2135 (df=6)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_ld_red) - logLik(mod_ld)),4)
```

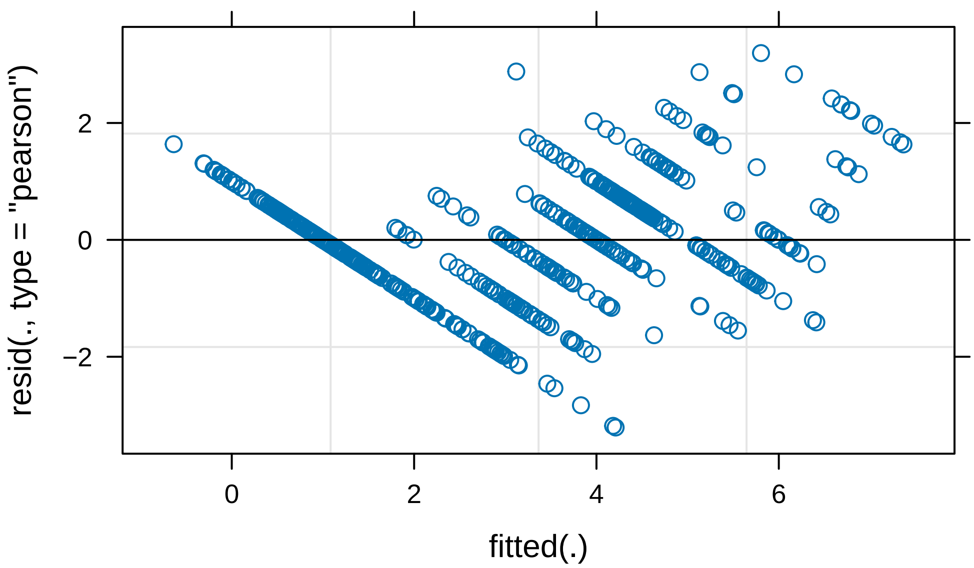
```
'log Lik.' 0 (df=6)
```

```r
summary(mod_ld)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
   Data: scott_long %>% filter(BLMD == "LD")

     AIC      BIC   logLik deviance df.resid
  2024.8   2068.8  -1002.4   2004.8      593

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.93811 -0.53078  0.01785  0.54850  2.92456

Random effects:
 Groups   Name        Variance Std.Dev.
 id       (Intercept) 0.6099   0.781
 Residual             1.1937   1.093
Number of obs: 603, groups:  id, 201

Fixed effects:
                   Estimate Std. Error t value
(Intercept)       4.364e-01  3.400e-01   1.284
age               1.216e-02  9.261e-03   1.313
I(age^2)          3.375e-05  5.428e-05   0.622
```

```
I(age^3)         -1.077e-07  8.860e-08  -1.215
sexmale          -1.121e+00  7.880e-01  -1.423
age:sexmale       3.002e-02  2.291e-02   1.310
I(age^2):sexmale -8.021e-05  1.541e-04  -0.520
I(age^3):sexmale  5.577e-08  2.965e-07   0.188

Correlation of Fixed Effects:
            (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age         -0.943
I(age^2)     0.912 -0.991
I(age^3)    -0.879  0.967 -0.992
sexmale     -0.431  0.407 -0.394  0.379
age:sexmale  0.381 -0.404  0.401 -0.391 -0.957
I(g^2):sxml -0.321  0.349 -0.352  0.349  0.925 -0.989
I(g^3):sxml  0.263 -0.289  0.296 -0.299 -0.886  0.961 -0.990
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

```r
plot(mod_ld)
```



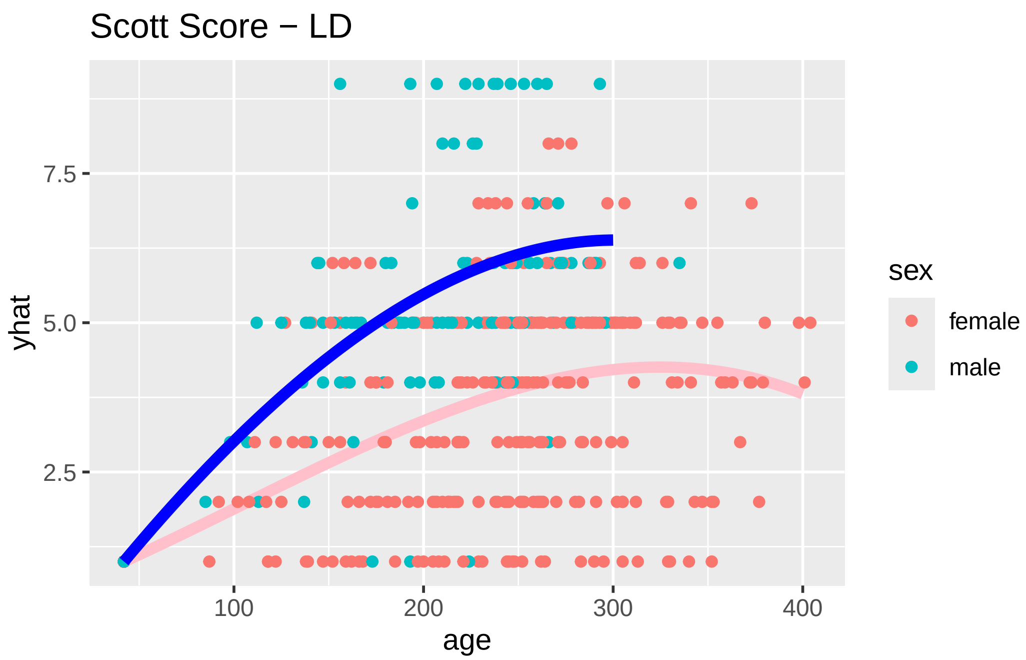

```r
confint(mod_bm)
```

```
Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)
```

```
Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation
```

```
                        2.5 %        97.5 %
.sig01             3.641796e-01  5.241933e-01
.sigma             6.138803e-01  7.053054e-01
(Intercept)       -2.663009e+00 -1.868469e+00
age                8.033269e-02  1.019677e-01
I(age^2)          -3.881395e-04 -2.614562e-04
I(age^3)           2.823800e-07  4.890441e-07
sexmale           -2.152828e+00 -3.115144e-01
age:sexmale        1.158228e-02  6.509252e-02
I(age^2):sexmale -4.217848e-04 -6.183189e-05
I(age^3):sexmale  1.092840e-07  8.017311e-07
```

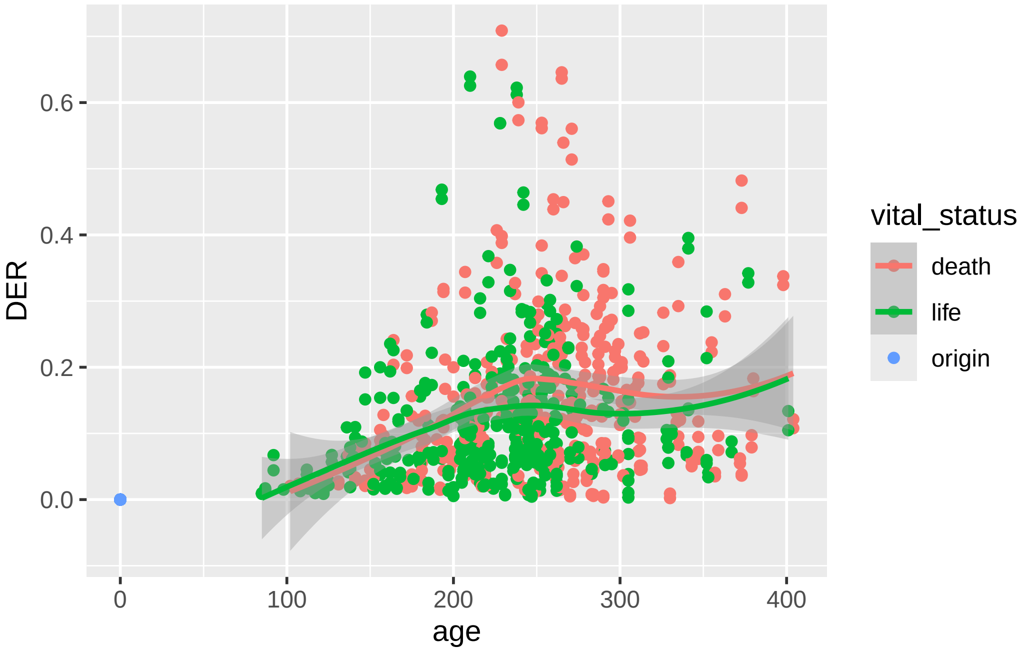
```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_ld, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_ld, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(a
```



Dentin Ratio

```
ggplot(aes(x = age, y = DER, color = vital_status), data = dentin_long) + geom_point() + geor
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
library(lme4)
mod_dentin_full <- lmer(DER ~ (age+I(age^2)+ I(age^3))*sex + (1|id) + (1|rater), data = dent:
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
mod_dentin_red <- lmer(DER ~ (age+I(age^2)+ I(age^3)) + (1|id) + (1|rater), data = dentin_lo1
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
#Test stat
-2*(logLik(mod_dentin_red) - logLik(mod_dentin_full))
```

```
'log Lik.' 137.6153 (df=7)
```

```r
#p-value
1 - pchisq(-2*(logLik(mod_dentin_red) - logLik(mod_dentin_full)),4)
```

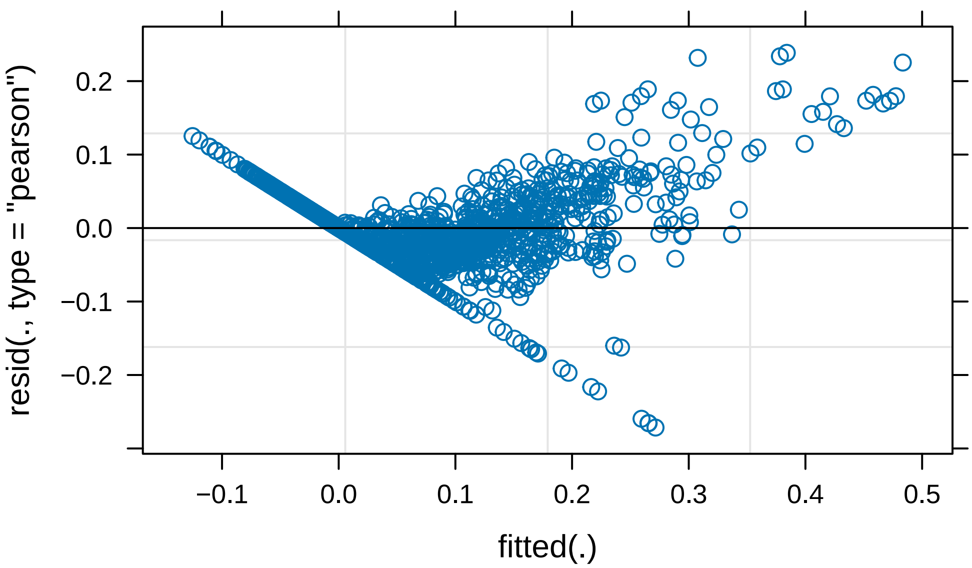
```
'log Lik.' 0 (df=7)
```

```r
summary(mod_dentin_full)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: DER ~ (age + I(age^2) + I(age^3)) * sex + (1 | id) + (1 | rater)
   Data: dentin_long

     AIC      BIC   logLik deviance df.resid
 -2974.3  -2918.2   1498.1  -2996.3     1195

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.6371 -0.5037 -0.0729  0.4644  4.0732

Random effects:
 Groups   Name        Variance  Std.Dev.
 id       (Intercept) 4.145e-03 0.064382
 rater    (Intercept) 1.709e-05 0.004134
 Residual             3.429e-03 0.058556
Number of obs: 1206, groups:  id, 201; rater, 2

Fixed effects:
                   Estimate Std. Error t value
(Intercept)      -3.198e-05  6.899e-03  -0.005
age              -2.793e-04  2.329e-04  -1.199
I(age^2)          5.667e-06  1.692e-06   3.350
I(age^3)         -1.009e-08  3.031e-09  -3.327
sexmale          -1.015e-04  1.241e-02  -0.008
age:sexmale      -5.171e-04  5.299e-04  -0.976
I(age^2):sexmale  7.898e-06  4.524e-06   1.746
I(age^3):sexmale -1.632e-08  9.639e-09  -1.693

Correlation of Fixed Effects:
            (Intr) age    I(g^2) I(g^3) sexmal ag:sxm I(^2):
age         -0.094
```

```
I(age^2)      0.053 -0.986
I(age^3)     -0.039  0.952 -0.989
sexmale      -0.456  0.052 -0.030  0.021
age:sexmale   0.041 -0.440  0.433 -0.419 -0.101
I(g^2):sxml -0.020  0.368 -0.374  0.370  0.055 -0.982
I(g^3):sxml  0.012 -0.299  0.311 -0.314 -0.039  0.942 -0.988
fit warnings:
Some predictor variables are on very different scales: consider rescaling
```

```r
plot(mod_dentin_full)
```

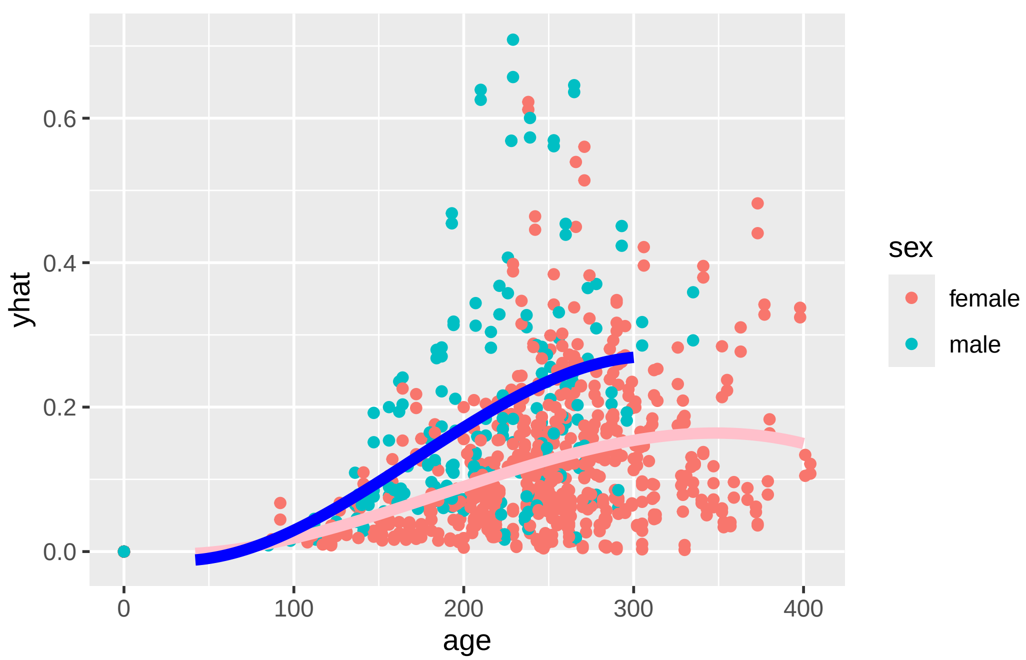



```r
confint(mod_dentin_full)
```

```
Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3):sexmale

Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation
```

```
Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3):sexmale: falling back to linear interpolation

                         2.5 %        97.5 %
.sig01              5.769599e-02  7.213582e-02
.sig02              0.000000e+00  2.170873e-02
.sigma              5.608466e-02  6.121426e-02
(Intercept)        -1.511011e-02  1.504710e-02
age                -7.361188e-04  1.775428e-04
I(age^2)            2.349205e-06  8.985359e-06
I(age^3)           -1.603070e-08 -4.139735e-09
sexmale            -2.449331e-02  2.429104e-02
age:sexmale        -1.556596e-03  5.223088e-04
I(age^2):sexmale -9.762482e-07  1.677352e-05
I(age^3):sexmale -3.523132e-08  2.590469e-09
```

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_dentin_full, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_dentin_full, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) +
  geom_point(aes(x = age, y = DER, col = sex), data = dentin_long) + geom_path(col = "pink",
```



```
##ICC analysis
## Only use death and cast ages
mod_dentin_full <- lmer(DER ~ (age+I(age^2)+ I(age^3))*sex + (1|id) + (1|rater), data = denti
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
mod_dentin_red <- lmer(DER ~ (age+I(age^2)+ I(age^3)) + (1|id) + (1|rater), data = dentin_lon
```

```
Warning: Some predictor variables are on very different scales: consider
rescaling
```

```
(VarCorr(mod_dentin_full)$rater[1])/(VarCorr(mod_dentin_full)$rater[1]+attr(VarCorr(mod_denti
```

```
[1] 0.001688996
```

```
(VarCorr(mod_dentin_full)$id[1])/(VarCorr(mod_dentin_full)$id[1]+attr(VarCorr(mod_dentin_full
```

```
[1] 0.2718641
```

Are there any patterns for rate of wear? That is, is there a standard wear through time (by month or year), or are some baboons wearing their teeth down faster that what we might expect?

Pava

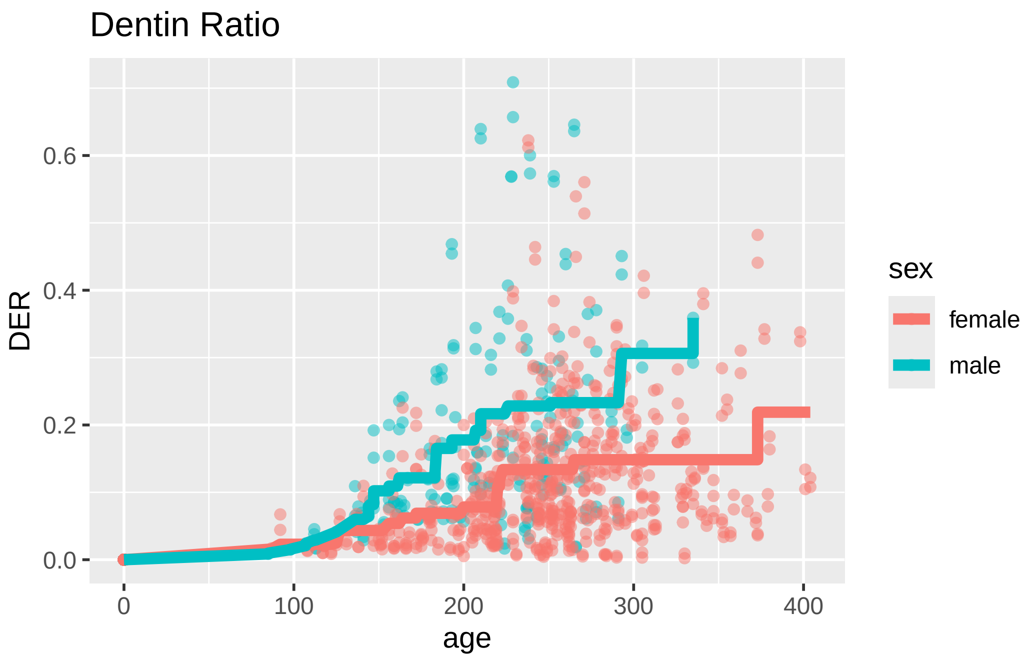
Dentin ratio

```
#Overall
library(Iso)
```

```
Iso 0.0-21
```

```
An "infelicity" in the function ufit() (whereby
it was all too easy to conflate the location of
the mode with its index in the entries of the
"x" argument) has been corrected.  To this end,
ufit() now has arguments "lmode" (the location
of the mode), and "imode" (its index).  At most
one of these arguments should be specified.  See
the help for ufit().
```

```
#Isotonic regression for BM.
#Separate curves for male and female.
sub <- dentin_long %>% filter(sex == "male") %>% arrange(age)
y <- dentin_long %>% filter( sex == "male") %>% arrange(age) %>% pull(DER)
sub$isoy <- pava(y)
submale <- sub

sub <- dentin_long %>% filter( sex == "female") %>% arrange(age)
y <- dentin_long %>% filter(sex == "female") %>% arrange(age) %>% pull(DER)
sub$isoy <- pava(y)
subfemale <- sub

sub_dentin <- rbind(submale, subfemale)

ggplot(aes(x = age, y = DER, color = sex), data = sub_dentin) + geom_point(alpha = 0.5) + geo
```



Scott scores

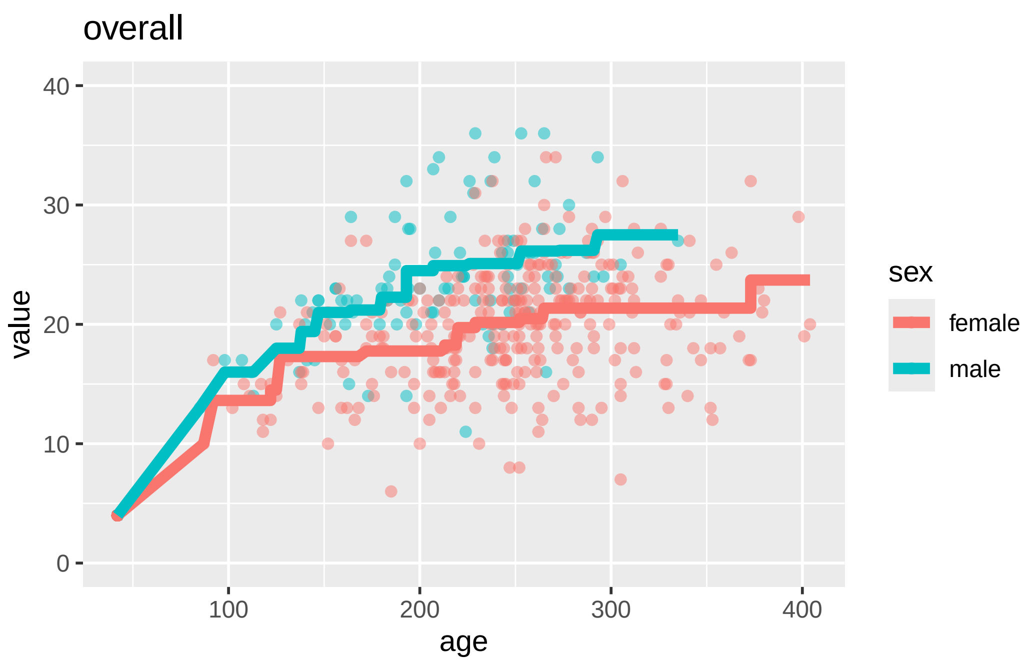
Scott scores summed across locations

```
#Overall
library(Iso)

#Isotonic regression for overall
scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(va
```

```
`summarise()` has grouped output by 'id', 'castost', 'age'. You can override
using the `.groups` argument.
```

```
#Separate curves for male and female.
sub <- scott_long_sum %>% filter(sex == "male") %>% arrange(age)
y <- scott_long_sum %>% filter(sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long_sum %>% filter(sex == "female") %>% arrange(age)
y <- scott_long_sum %>% filter(sex == "female") %>% arrange(age) %>% pull(value)
```

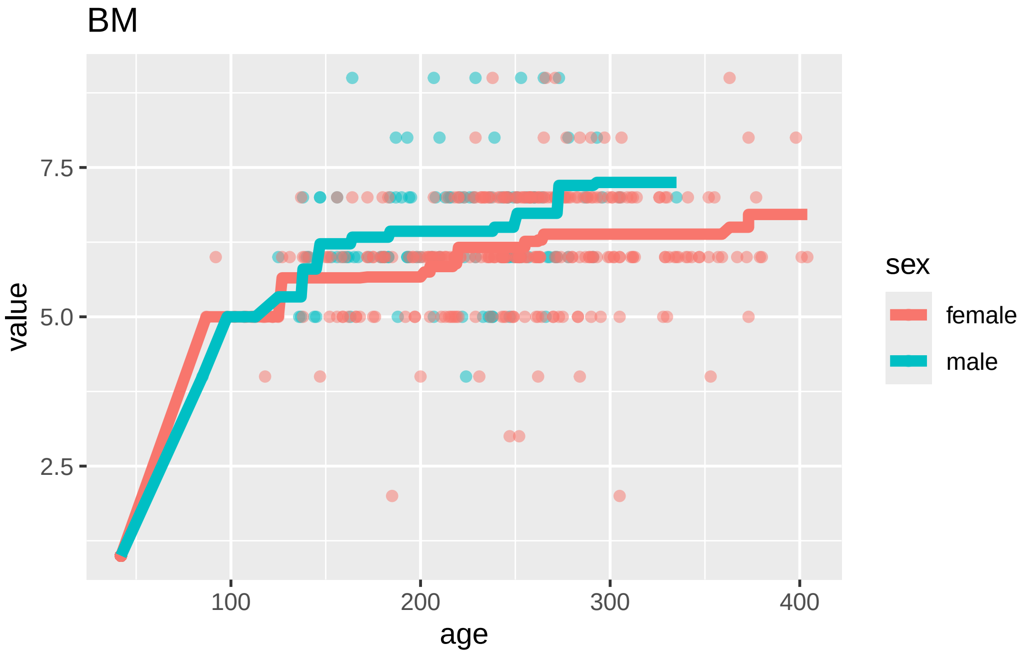
```
sub$isoy <- pava(y)
subfemale <- sub

suball <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = suball) + geom_point(alpha = 0.5) + geom_
```



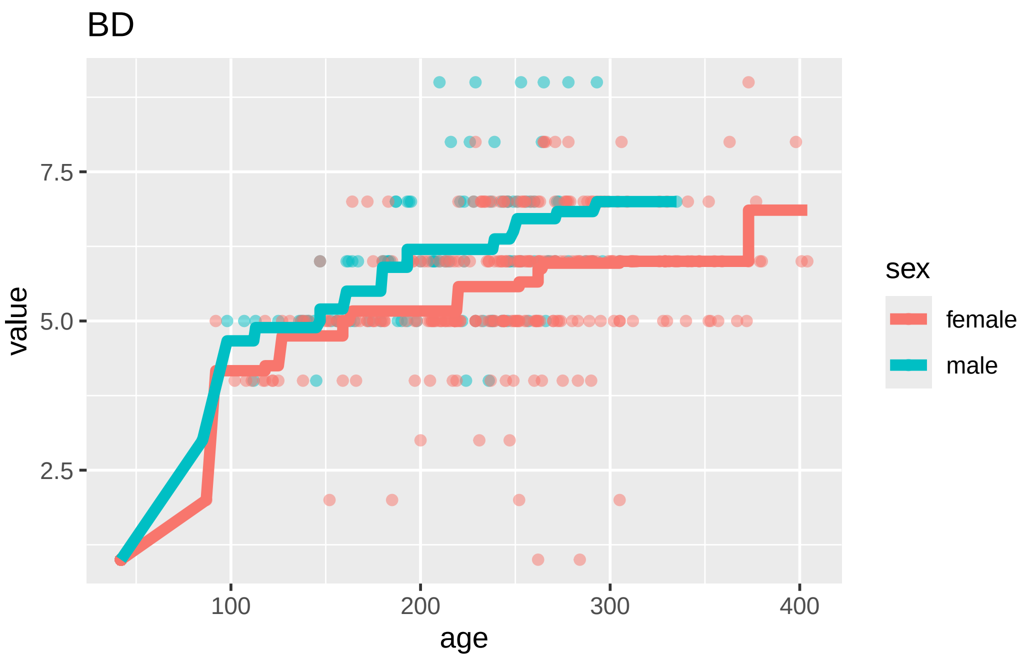
Individual locations

```
library(Iso)

#Isotonic regression for BM.
#Separate curves for male and female.
type <- "BM"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
```

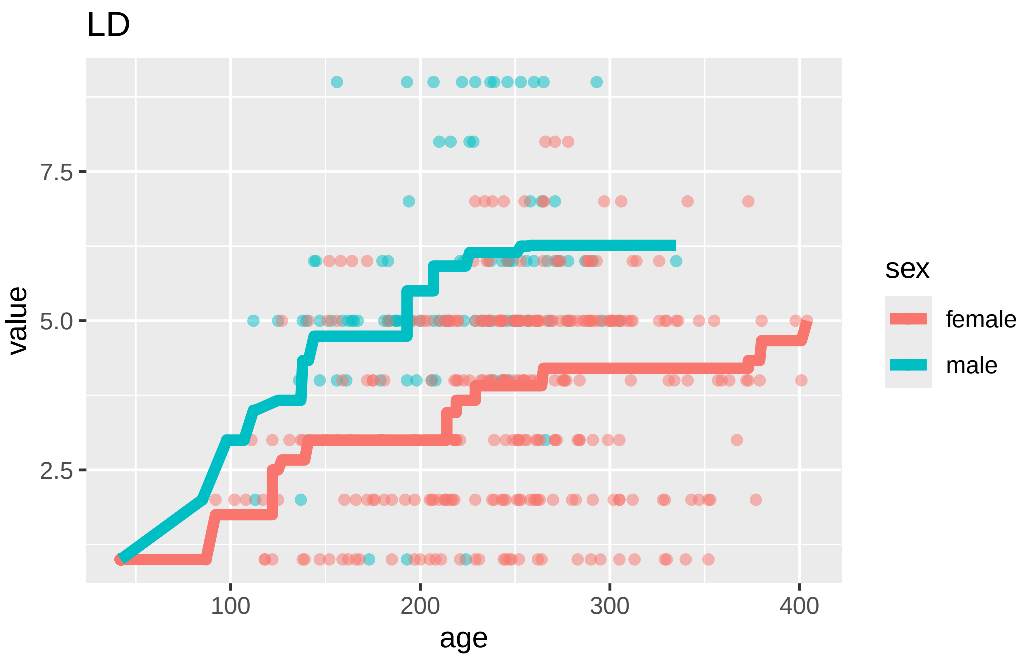
```
sub$isoy <- pava(y)
subfemale <- sub

subBM <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subBM) + geom_point(alpha = 0.5) + geom_
```



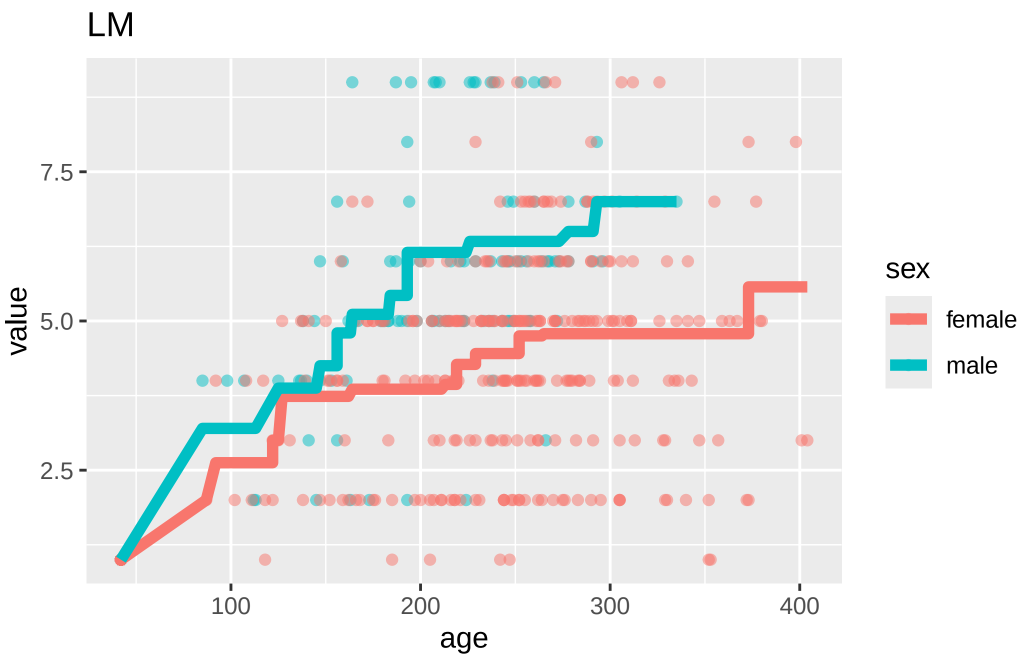

```
#Isotonic regression for BD.
#Separate curves for male and female.
type <- "BD"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subBD <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subBD) + geom_point(alpha = 0.5) + geom_
```




```
#Isotonic regression for LD.
#Separate curves for male and female.
type <- "LD"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subLD <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subLD) + geom_point(alpha = 0.5) + geom_p
```




```
#Isotonic regression for LM.
#Separate curves for male and female.
type <- "LM"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subLM <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subLM) + geom_point(alpha = 0.5) + geom_p
```




```
suball <- rbind(subBM, subBD, subLM, subLD)

# ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball,alpha = 0.5) + geo

suball <- suball %>% mutate(BLword = case_when(BL == "B" ~ "Buccal",
                                               BL == "L" ~ "Lingual"),
                            MDword = case_when(MD == "M" ~ "Mesial",
                                               MD == "D" ~ "Distal"))



pdf(file = "/Users/gregorymatthews/Dropbox/baboon_git/isotonic.pdf", h = 5, w = 5, pointsize
ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball, size = 0.5) + geom_
  values = c("#F8766D","#00BFC4"),
  labels = c( "Female","Male")
)
dev.off()
```

```
pdf
  2
```

```
pdf(file = "/Users/gregorymatthews/Dropbox/baboon_git/isotonic_qword.pdf", h = 5, w = 5, poin
ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball,size = 0.5) + geom_
  values = c("#F8766D","#00BFC4"),
```

```
  labels = c( "Female","Male")
)
dev.off()
```

```
pdf
  2
```

```
#
# type <- "BD"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$isoy <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso
#
# type <- "LM"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$isoy <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso
#
# type <- "LD"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$isoy <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso
```

#Comparing scott and DER

```
#sum the scott scorres
#Average the dentin across raters.
#merge the scottt and DER data
#Plot scott vs DER.

scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(va
```

```
`summarise()` has grouped output by 'id', 'castost', 'age'. You can override
using the `.groups` argument.
```

```
dentin_long_sum <- dentin_long %>% group_by(id, age) %>% summarize(DER = mean(DER))
```

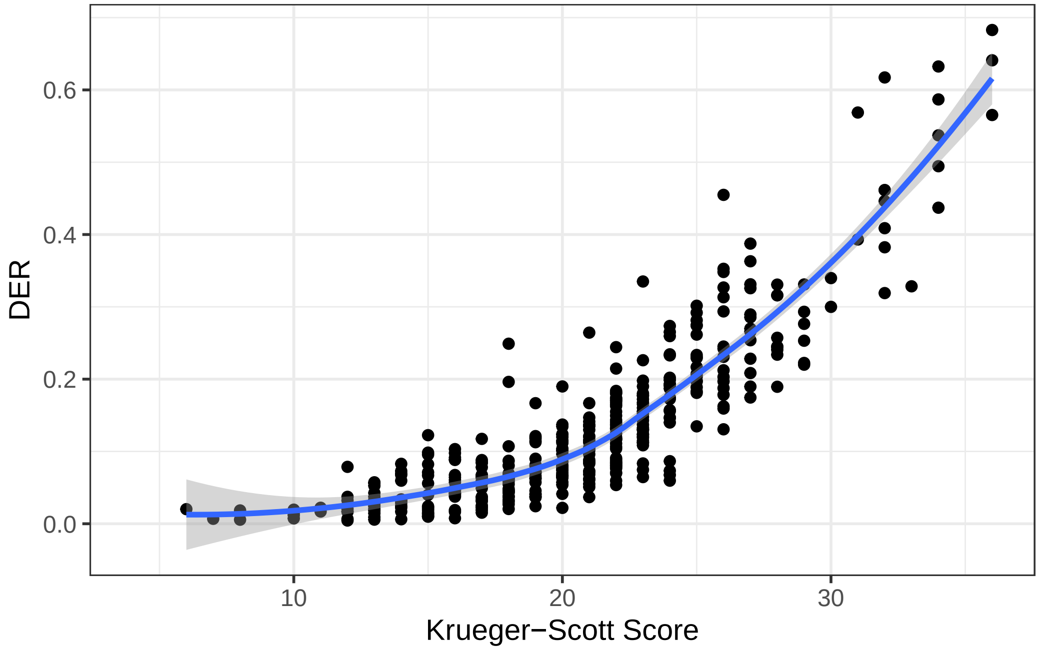
```
`summarise()` has grouped output by 'id'. You can override using the `.groups`
argument.
```

```
merg <- scott_long_sum %>% left_join(dentin_long_sum, by = c("id", "age"))

ggplot(aes(x = value, y = DER),data = merg) + geom_point() + xlab("Krueger-Scott Score") + ge
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 201 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

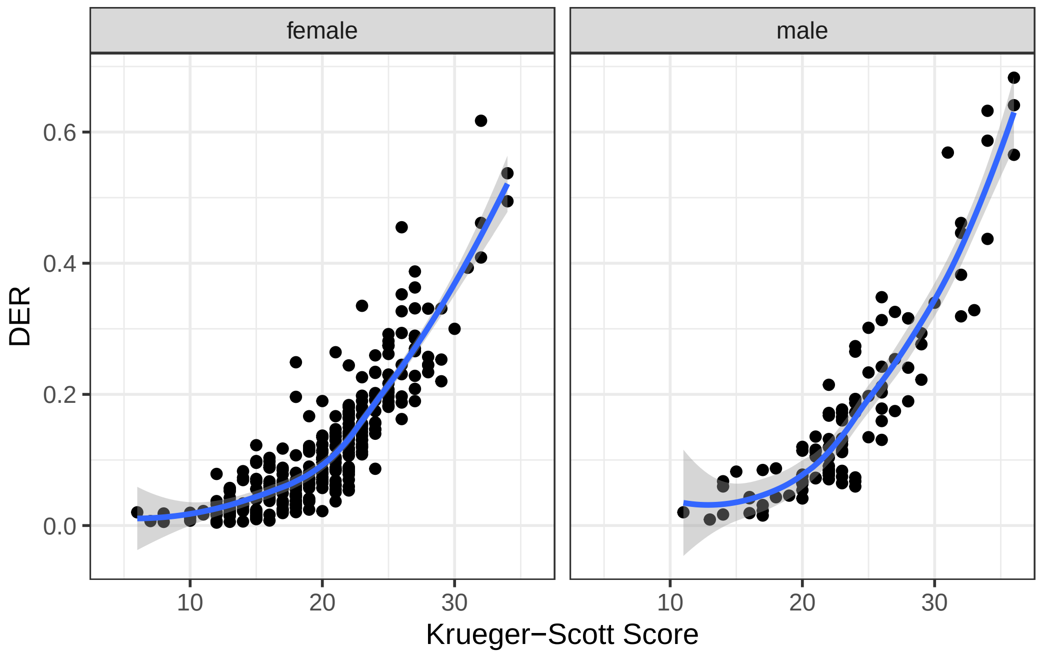
```
Warning: Removed 201 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
ggplot(aes(x = value, y = DER),data = merg) + geom_point() + xlab("Krueger-Scott Score") + ge
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 201 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 201 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Notes for Greg

- https://link.springer.com/article/10.1007/s11222-017-9797-8
- https://academic.oup.com/jrsssb/article/83/5/963/7056107